Take Home Exam I
Key

1. $G = B(0,1) \setminus \{[-1,-\tfrac{1}{2}] \cup [\tfrac{1}{2}i, i] \cup [\tfrac{1}{2}, 1]\}$

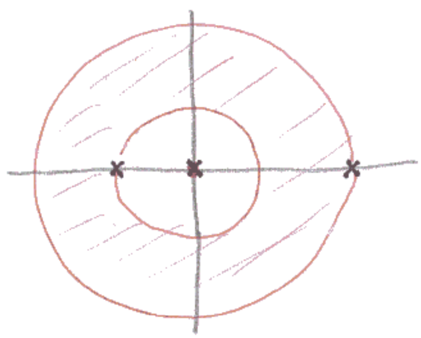
2. a) $G = B(0,1)$

   b) $G = B(0,1) \cup B(3,1)$

   c) $G = \mathbb{C} \setminus \{0\}$

3. $f(z) = \dfrac{3}{z+1} + \dfrac{2}{z-2} - \dfrac{4}{z}$

a) $\operatorname{ann}(0; 1, 2)$

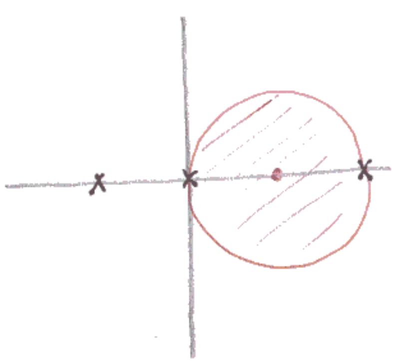


$$\frac{3}{z+1} + \frac{2}{z-2} - \frac{4}{z}$$

$$= \frac{3}{z}\,\frac{1}{1+\frac{1}{z}} - \frac{2}{2}\,\frac{1}{1-\frac{z}{2}} - \frac{4}{z}$$

$$= \frac{3}{z}\sum_0^\infty \left(-\frac{1}{z}\right)^n - \sum_0^\infty \left(\frac{z}{2}\right)^n - \frac{4}{z}$$

b) $\operatorname{ann}(1; 0, 1)$



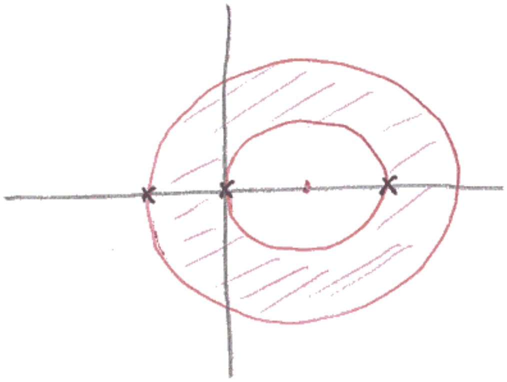
$$\frac{3}{z+1} + \frac{2}{z-2} - \frac{4}{z}$$

$$= \frac{3}{2+(z-1)} + \frac{2}{-1+(z-1)} - \frac{4}{1+(z-1)}$$

$$= \frac{3}{2}\,\frac{1}{1+\frac{z-1}{2}} - \frac{2}{1-(z-1)} - \frac{4}{1+(z-1)}$$

$$= \frac{3}{2}\sum_0^\infty \left(-\frac{z-1}{2}\right)^n - 2\sum_0^\infty (z-1)^n - 4\sum_0^\infty [-(z-1)]^n$$

c) $\operatorname{ann}(1; 1, 2)$



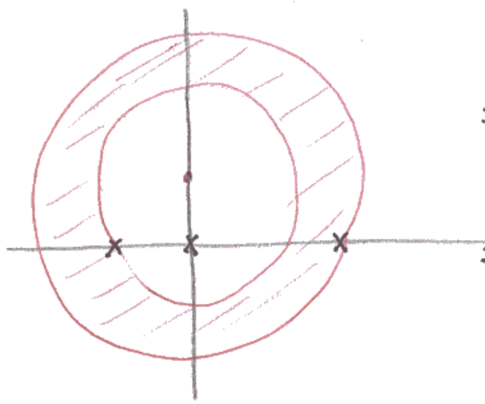
$$\frac{3}{z+1} + \frac{2}{z-2} - \frac{4}{z}$$

$$= \frac{3}{2+(z-1)} + \frac{2}{-1+(z-1)} - \frac{4}{1+(z-1)}$$

$$= \frac{3}{2}\,\frac{1}{1+\frac{z-1}{2}} + \frac{2}{z-1}\,\frac{1}{1-\frac{1}{z-1}} - \frac{4}{z-1}\,\frac{1}{1+\frac{1}{z-1}}$$

$$= \frac{3}{2}\sum_0^\infty\left(-\frac{z-1}{2}\right)^n + \frac{2}{z-1}\sum_0^\infty\left(\frac{1}{z-1}\right)^n - \frac{4}{z-1}\sum_0^\infty\left(-\frac{1}{z-1}\right)^n$$

d) $\operatorname{ann}(i; \sqrt{2}, \sqrt{5})$



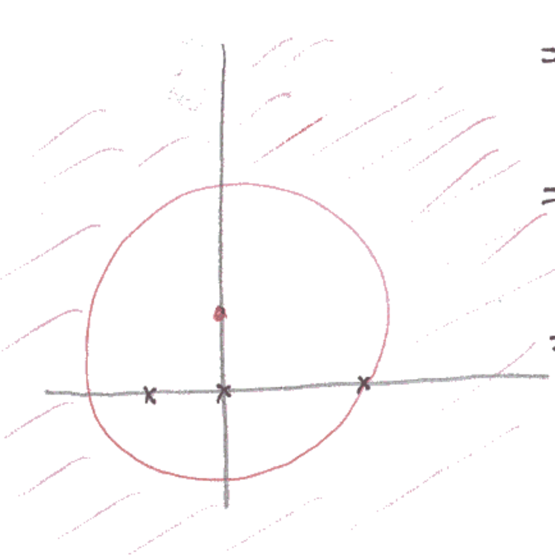
$$\frac{3}{z+1} + \frac{2}{z-2} - \frac{4}{z}$$

$$= \frac{3}{1+i+(z-i)} + \frac{2}{-2+i+(z-i)} - \frac{4}{i+(z-i)}$$

$$= \frac{3}{z-i}\,\frac{1}{1+\frac{1+i}{z-i}} + \frac{2}{-2+i}\,\frac{1}{1+\frac{z-i}{-2+i}} - \frac{4}{z-i}\,\frac{1}{1+\frac{i}{z-i}}$$

$$= \frac{3}{z-i}\sum_0^\infty\left(-\frac{1+i}{z-i}\right)^n + \frac{2}{-2+i}\sum_0^\infty\left(-\frac{z-i}{-2+i}\right)^n - \frac{4}{z-i}\sum_0^\infty\left(-\frac{i}{z-i}\right)$$

e) $\operatorname{ann}(i; \sqrt{5}, \infty)$



$$\frac{3}{z+1} + \frac{2}{z-2} - \frac{4}{z}$$

$$= \frac{3}{1+i+(z-i)} + \frac{2}{-2+i+(z-i)} - \frac{4}{i+(z-i)}$$

$$= \frac{3}{z-i}\,\frac{1}{1+\frac{1+i}{z-i}} + \frac{2}{z-i}\,\frac{1}{1+\frac{-2+i}{z-i}} - \frac{4}{z-i}\,\frac{1}{1+\frac{i}{z-i}}$$

$$= \frac{3}{z-i}\sum_0^\infty\left(-\frac{1+i}{z-i}\right)^n + \frac{2}{z-i}\sum_0^\infty\left(-\frac{-2+i}{z-i}\right)^n - \frac{4}{z-i}\sum_0^\infty\left(-\frac{i}{z-i}\right)$$

4a) $z=0$ pole order 2 $S_f(0) = -\frac{1}{z^2}$

$z=1$ pole order 1

$z=-1$ pole order 1

$z=i$ removable define $f(i) = \lim_{z\to i} f(z) = \frac{1}{2}$

$z=-i$ removable define $f(-i) = \lim_{z\to -i} f(z) = \frac{1}{2}$

b) $z=0$ pole order 1 $S_f(0) = \frac{1}{iz}$

$z = 2\pi k$, $k \in \mathbb{Z}^+$, pole order 1

c) $z=0$ essential singularity

$z=1$ pole order 1

$z=-1$ pole order 1

$z=i$ pole order 2

$z=-i$ pole order 2

5a. $$f(z)= \frac{z^3\left(2z^3 - \frac{(2z^3)^3}{3!} + \cdots\right)}{\left(1-\frac{(z^4)^2}{2!}+\cdots\right)-1} = \frac{z^6\left(2-\frac{1}{3}z^6+\cdots\right)}{z^8\left(-\frac{1}{2}+\frac{z^8}{4!}+\cdots\right)} = \frac{1}{z^2}g(z)$$

pole order 2 $\operatorname{Res}(f,0)=0$ since g is even

b) $$f(z) = \frac{z^3\left(z-\frac{z^3}{3!}+\cdots\right)^3}{\left[1-\left(1-\frac{(z^2)^2}{2!}+\cdots\right)\right]^2} = \frac{z^6}{z^8}\frac{\left(1-\frac{1}{2}z^2+\cdots\right)}{\left(\frac{1}{4}-\frac{1}{24}z^4+\cdots\right)} = \frac{1}{z^2}g(z)$$

pole order 2 $\operatorname{Res}(f,0)=0$ since g is even

6. Let γ be simple, rectifiable curve in $\mathbb{C} \setminus \{-2, 2\}$ joining 0 to 1. Let λ be the direct line segment $[1, 0]$

Then $\sigma = \gamma \cup \lambda$ is a closed rectifiable curve

$$\int_\sigma \frac{dz}{4-z^2} = \int_\sigma \frac{\frac{1}{4}}{z+2}\,dz - \int_\sigma \frac{\frac{1}{4}}{z-2}\,dz$$

$$= \frac{2\pi i}{4}\left[n(\sigma, -2) - n(\sigma, 2)\right]$$

$$\therefore \quad \int_\gamma \frac{dz}{4-z^2} = \int_0^1 \frac{dx}{4-x^2} + \frac{2\pi i}{4}\left[n(\sigma, -2) - n(\sigma, 2)\right]$$

$$= \frac{\log 3}{4} + \frac{2\pi i}{4} k \quad , \quad k \in \mathbb{Z}$$